

Q.22 Write any four applications of inductive type proximity sensor.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 What is vibration? Explain the working of seismic transducer with diagram.

Q.24 What is Viscosity? What are the types of viscosity measurement techniques? Explain any one type of it with diagram.

Q.25 Explain the working of the following with diagram-

- a) pH meter
- b) Photo electric tachnogenerator

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Roll No.

6th Sem/ Instrumentation & Control Subject : Advance Measurement Technique

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 In a measurement system, the output is obtained on which of the following block?

- a) Primary sensing element
- b) Data representation element
- c) Data transmission element
- d) Data storage element

Q.2 Which of the following instrument is used for the measurement of area?

- a) Protractor
- b) Dial bevel protractor
- c) Numerical method
- d) Micro meter

Q.3 An stroboscope is used for the measurement of

- a) Linear velocity
- b) Angular velocity
- c) Weight
- d) Viscosity

- Q.4 Which of the following instruments is used to measure the thickness?
 a) Doppler Transducer b) LVDT
 c) Resistive transducer d) Potentiometer
- Q.5 Rotating cylinder method is used for the measurement of _____.
 a) Viscosity b) Velocity
 c) Vibrations d) pH of the solution
- Q.6 Which of the following proximity sensor is used to detect metallic objects
 a) Magnetic proximity sensor
 b) Inductive proximity sensor
 c) Ultrasonic proximity sensor
 d) Both a & b

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Vernier caliper is used for the ____ measurement.
- Q.8 A dial bevel protractor can measure angle up to 90° . (T/F)
- Q.9 Doppler velocity measurement method works on the principle of ____.

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- Q.10 A pH value of 9.2 shows acidic behaviour. (T/F)
- Q.11 Falling sphere method is used to measure ____.
- Q.12 Define proximity sensor.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the primary sensing elements of measurement system with example.
- Q.14 Explain any method for the measurement of angle with diagram.
- Q.15 How numerical method is used for the measurement of area?
- Q.16 Explain the working of Vernier caliper for the measurement of length.
- Q.17 Draw and explain the tooth rotor variable reluctance tachogenerator.
- Q.18 What is tachogenerator? How it is used to measure angular velocity?
- Q.19 Draw and explain the method of measurement of specific gravity.
- Q.20 Explain how a resistor is used to measure the thickness of an object.
- Q.21 Explain the working of falling piston viscometer.

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